What's Wrong With Science?

Defining the Future of Science

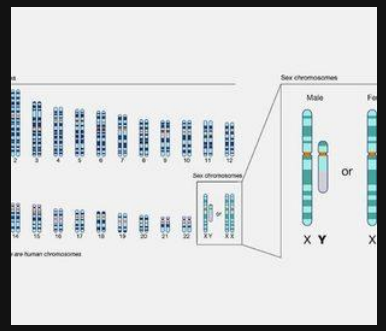
Session 7 · Professor David Gresham · Guest: Michael Purugganan, Dean, NYU Faculty of Arts & Science · October 21, 2026



Michael Purugganan

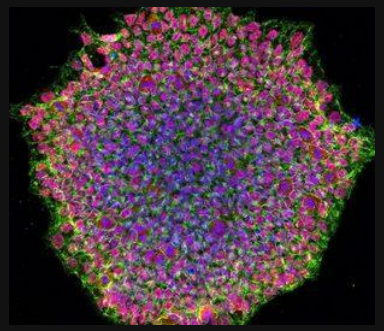
Silver Professor of Biology and the Dean of the Faculty of Arts and Science of New York University

25 years of breakthroughs



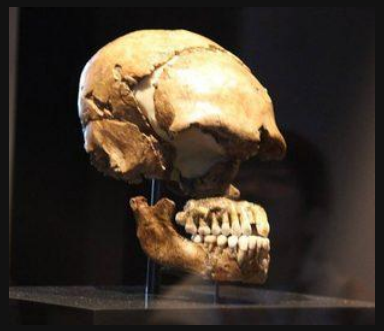
2001

The human genome



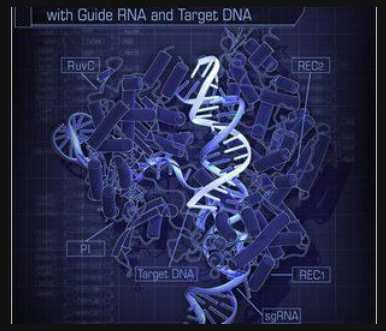
2006

Reprogramming adult cells into stem cells



2010

The Neanderthal genome



2012

CRISPR gene editing



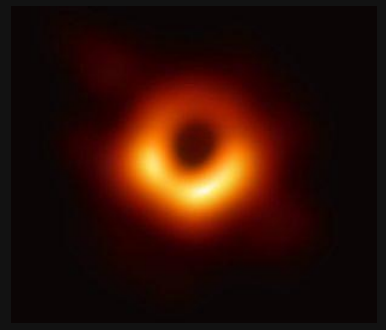
2015

Gravitational waves detected



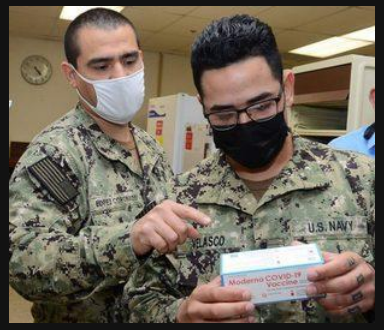
2017

GLP-1 drugs for diabetes and obesity



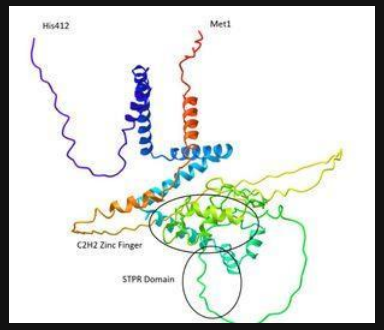
2019

First image of a black hole



2020

mRNA vaccines



2020

AlphaFold predicts protein structures

Every discovery began as a question nobody knew needed an answer



1993

Odd DNA repeats in salt-pond microbes, Spain →



1992

A hormone in Gila monster venom →



2005

How to stop cells attacking synthetic RNA →

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2020

mRNA COVID vaccines



The endless frontier (1945)

The breakthroughs came from the system Vannevar Bush designed:

- public money
- universities
- and curiosity-driven research



The endless frontier (1945)

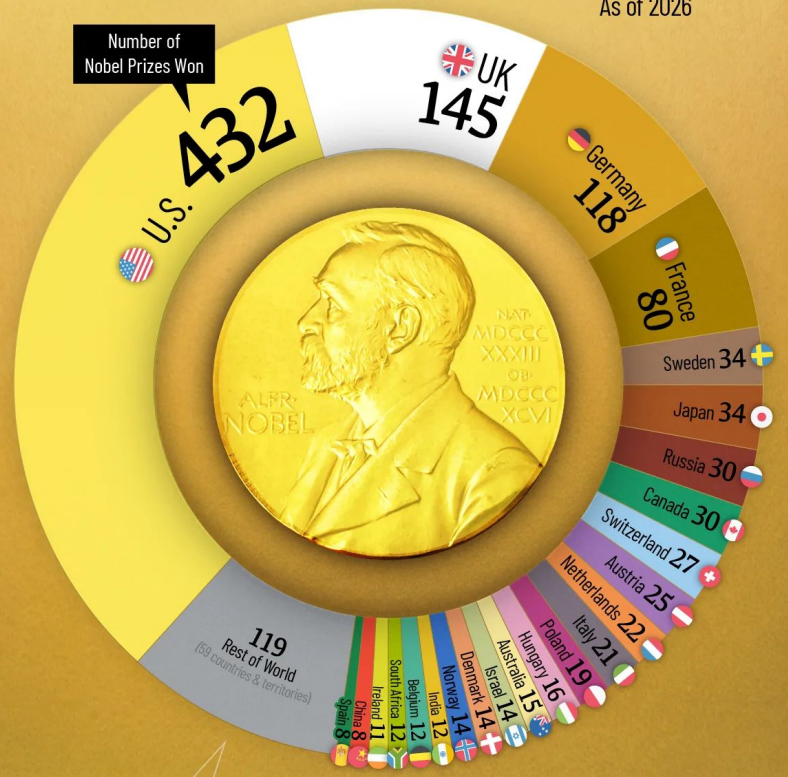
The breakthroughs came from the system Vannevar Bush designed:

- public money
- universities
- and curiosity-driven research

- A culture in the USA of welcoming international students and scientists

COUNTRIES WITH THE MOST NOBEL PRIZES

As of 2026



Number of Nobel Prizes Won

6: BLR CZE UKR

5: ARG EGY FIN

4: ROU TWN

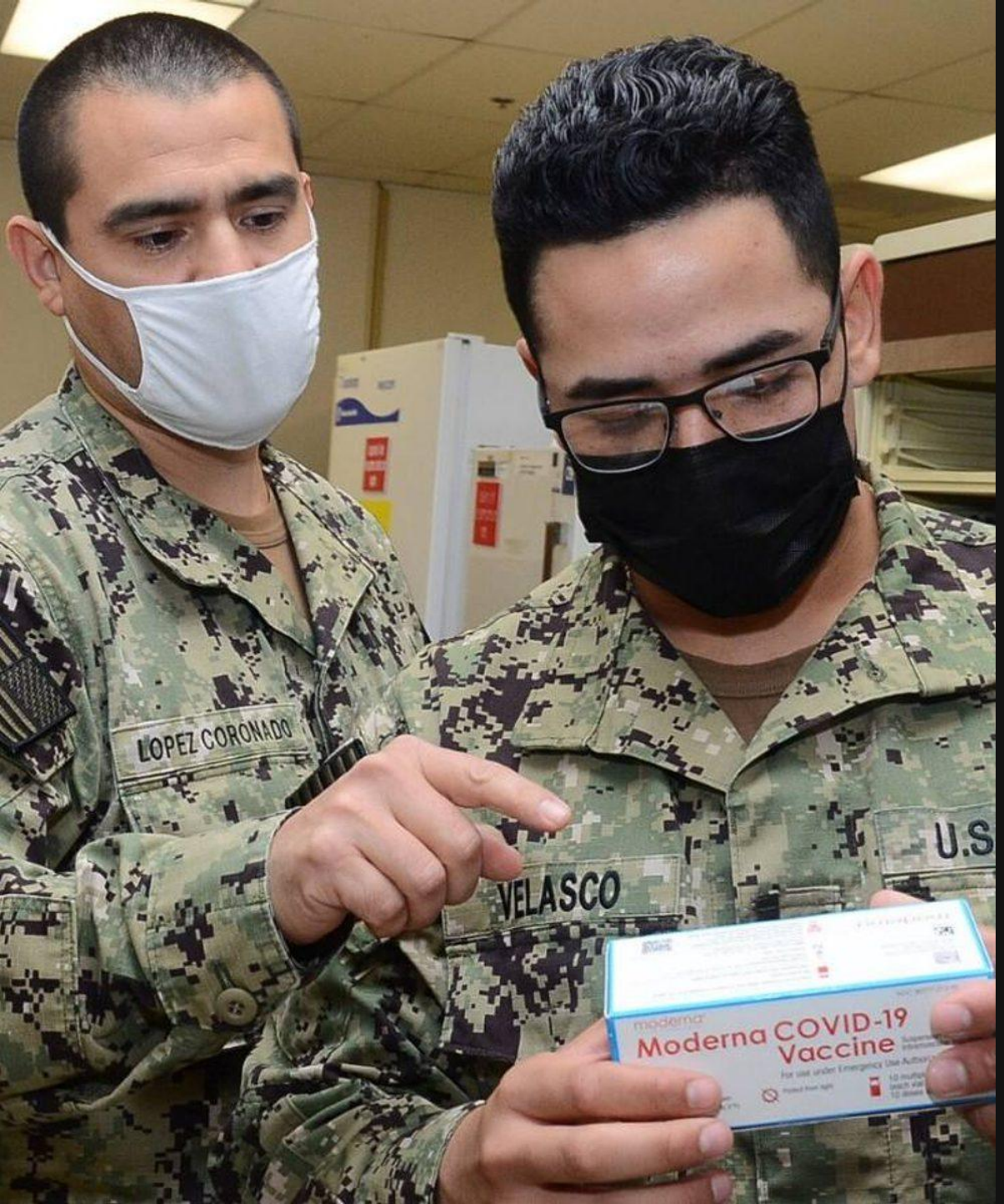
3: LTU MEX NZL KOR TUN TUR

2: DZA AZE CHL COL CYP GRC GTM IRN LBR LUX PAK PRT LCA TLS VEN

1: BGD BRA BGR CRI COD ETH FRO GHA HKG ISL IRQ JOR KEN LVA LBN MAR MMR NGA MKD PSE PER PHL SAU TZA Tibet TTO VNM YEM YUG ZWE

VISUAL CAPITALIST

Source: Wikipedia and Nobel Prize records, updated to October 2026.
Some laureates are associated with more than one country.



Where this course started

March 8, 2021

First dose of an mRNA vaccine, made in under a year.

Four years later:

- vaccine advisory committees dismantled
- mRNA vaccine funding halted
- research grants cut
- NSF, CDC, WHO defunded
- Increased mistrust
- Politicization of science

What happened?

Six weeks of “What’s wrong with science?”



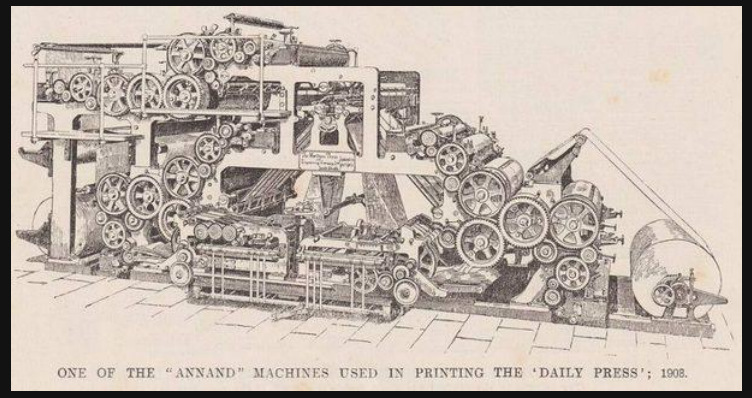
1 How science works

Michael Strevens, NYU



2 Who pays

Nate Williams, Sloan



3 How it's communicated

Apoorva Mandavilli, NYT



4 The dark side

Nina Gray, NYU



5 When to believe

Ivan Oransky, Retraction Watch & NYU



6 Trust and politics

Pat Egan, NYU

What is wrong with science?



What is wrong with science?

Nobody has responded yet.

Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What is the future of science?



100 scientists

“What is a big unanswered question in your field?”
-October 5, 2026

https://www.nytimes.com/2026/10/05/science/100-unanswered-questions-science.html

vs.

87 NYU students

“What is the most important unsolved problem in science?”
October 21, 2026

Which ones were posed by a student in this class?

(A) How does physical matter in the brain produce subjective conscious experience?

0%

(B) Why are humans the only species with a chin?

0%

(C) Are we alone in the universe?

0%

(D) Why do we sleep?

0%

(E) How can we make advances in science benefit everyone equally?

0%

(F) How can we prevent q day?

0%

(G) Do aliens exist?

0%

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Student

How does physical matter in the brain produce subjective conscious experience?

one of 12 of you who asked about consciousness (scientists: 1 of 100)

Scientist

Why are humans the only species with a chin?

Daniel Lieberman, evolutionary biologist, Harvard (NYT #37)

Both

Are we alone in the universe?

word for word: one of you, and Robert Hazen, Carnegie Institution for Science (NYT #99)

Scientist

Why do we sleep?

Katalin Karikó, Nobel laureate for mRNA vaccines (NYT #21)

Student

How can we make advances in science benefit everyone equally?

one of 14 of you who asked about how science works in society (scientists: 2)

Student

Do aliens exist? and How can we prevent q day?

Same curiosity, different verbs



Fix it

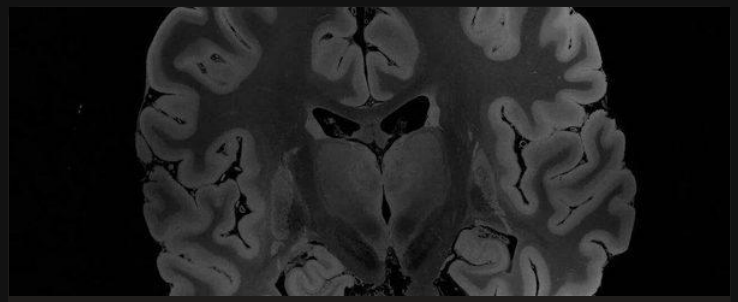
"How do we…?"

47% of students

14% of scientists

How will we slow down climate change?

Can we make the wildlife trade sustainable?



Understand it

"How does…? What is…?"

34% of students

53% of scientists

What is the nature of consciousness?

What is dark matter?



Reconstruct the past

"When did…? Who were…?"

6% of students

25% of scientists

How did life first emerge on Earth?

Who were the Sea Peoples?

Where the lists diverge most



Consciousness

12	1
students	scientists



Artificial intelligence

8	0
students	scientists



Curing cancer

7	3
students	scientists



Stopping aging

5	1
students	scientists



Origin of life

5	2
students	scientists



Life elsewhere

4	1
students	scientists



Dinosaurs

0	3
students	scientists



Ancient civilizations

0	4
students	scientists

Number of questions out of 87 students and 100 scientists. Climate change: 4 students, 4 scientists.

Example

My question

How can scientists overcome antibiotic resistance in bacteria such as MRSA?

Sharpened

Can we find new classes of antibiotics in soil microbes faster than bacteria evolve resistance to them?

2051 headline

"First new antibiotic class in decades ends the superbug era"

What had to happen?

Basic science

Soil microbiology; how resistance evolves

Money

Who pays for a drug that is rarely used?

Who and where

University labs, biotech, public funders

Trust and ethics

Activity: write the headline from 2051

1

Share

Share your pre-class question with your whole row.

Yes, you can change your question.

2

Sharpen

Turn it into a question someone could actually start working on.

3

Headline

Write the 2051 headline that says it was solved.

4

Post

Post your headline on poll everywhere.

Upvote the best ones

Post your headline (and upvote the best ones)

Nobody has responded yet.

Hang tight! Responses are coming in.

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Thank you for seven remarkable weeks!!!!



IN CONVERSATION

Michael Purugganan

Dean, NYU Faculty of Arts & Science · Professor of Biology

5-minute break